

Assignment - 1

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Operating system (os) and Types of os
Operating system :-

An operating system is a system software that act as an interface between the user and the computer hardware. It manage hardware resource execute programs control memory, file, input. output devices and provide a platform for application software to run efficiently.

① Batch Batch Operating system :-

A Batch OS is an operating system that controls similar jobs or tasks into groups called batches and processes them one after another automatically. The user does not interact with the OS. completely all the jobs in sequence without manual intervention. It is mainly used for repetitive and large-scale processing tasks.

* Advantages

- * Efficient for processing
- * Reduce CPU idle time
- * very less user intervention
- * suitable for repetitive tasks
- * Improve overall system utilization

* Disadvantages

- * No direct interaction bⁿ user & system
- * long waiting time
- * Difficult to identify & correct error
- * urgent jobs cannot execute immediately
- * Not suitable for interactive application

* Usage :-

- * payroll processing
- * Bank statement generation
- * Electricity bill generation
- * Examination result processing
- * Large data processing

2 Multiprogramming Operating System :-

A multiprogramming OS keeps multiple programs in the computer's main memory at the same time. When one program is waiting for an output or input operation, the CPU immediately switches to another program instead of remaining idle. This improves CPU utilization, increases system efficiency, and allows several programs to be executed simultaneously.

* Advantages :-

- * Maximize CPU utilization
- * Reduce CPU idle time
- * Improve system performance
- * Increase overall throughput
- * allow multiple programs to run simultaneously.

* Disadvantages:

- * Complex memory management
- * Requires efficient scheduling algorithms
- * Difficult to debug & maintain
- * May lead to deadlock problem.

* Usage

- * mainframe computer
- * Business applications
- * scientific computing
- * Enterprise system
- * Multi-user environments

3 multitasking operating system:

A multitasking OS allows multiple users or multiple program to use the computer system at the same time. It divides the CPU time into small intervals called time slices and allocates them to different tasks. This creates the impression that all program are running simultaneously while ensuring quick response and efficient utilization of system resources.

* Advantages

- * Fast Responsive time
- * Allows multiple application to run simultaneously
- * Improve user productivity
- * Efficient utilization of CPU Resources
- * supports multiple users.

* Disadvantages

- * Requires more memory
- * performance decrease if many program simultaneously
- * security issue may rise
- * more complex than single task OS

* Usage

- * windows OS
- * Linux OS
- * mac OS
- * office computers
- * Educational institutions.

4 Multiprocessor Operating System :-

A multi-processor that shares the same memory and other processor that shares the same memory and other system resource. The OS distributes tasks among the processor can run simultaneously. This increases processing speed improve system performance & provides greater reliability by efficiently utilizing all available processors.

* Advantages :-

- * Faster execution of programs
- * High system reliability
- * Better resource sharing
- * supports parallel processing
- * Increase overall throughput.

* Disadvantages:-

- * Expensive hardware
- * complex operating system design
- * Difficult maintenance
- * Higher implementation cost

* Usage

- * Super Computers
- * Data centers
- * Artificial Intelligence Applications
- * scientific research
- * Engineering simulations.

5 Real-Time operating system (RTOS)

A Real-Time OS is designed to process data and respond to event with specific time limit. It ensures that important tasks are completed without delay making it suitable for time-critical applications. RTOS provides predictable performance and is commonly used in system where even a small delay can lead to severe consequences.

* Advantages

- * very fast response time
- * Highly reliable
- * Accurate task execution
- * stable and predictable performance
- * suitable for critical applications.

* Disadvantage

- * Expensive to develop
- * Limited multitasking capability
- * Complex programming & debugging
- * Requires specialized hardware in some cases

* Uses

- * Air traffic control system
- * Medical equipment
- * Robotics
- * Industrial automation
- * Automobile control system

6. Distributed operating system.

A Distributed OS connects multiple independent computers through a network and manages them as a single integrated system. It allows users to share resources, data and processing efficiently. The OS coordinates communication bⁿ different computer making the entire network appear as one unified system.

* Advantages

- * Efficient resource sharing
- * High reliability
- * Easy system expansion
- * Better performance
- * Provide fault tolerance.

* Disadvantages:-

- * Depends on network connectivity
- * security challenges
- * complex system management
- * High installation cost
- * Difficult troubleshooting

* Usage

- * cloud computing
- * Banking system
- * Research laboratories
- * Large organization
- * Distributed database

7. * Network operating system (NOS)

A Network OS manage and control computer connection through a network it allow user to share file printers application and other network resource security. It also provides centralized management user authentication security and communication between different computer connected to the same network

* Advantages

- * centralized network management
- * Easy file and printer sharing
- * strong security features
- * Easy backup & maintenance
- * support multiple user efficiently

* Disadvantages.

- * server failure can affect all users
- * High installation and maintenance cost
- * Requires skilled network administrators
- * Regular maintenance is necessary
- * Network congestion may reduce performance

* Usage

- * school and colleges
- * offices
- * Banks
- * Hospitals
- * corporate organization.